

QUANTITATIVE RESEARCH NOTE: POST-CRASH MEAN REVERSION IN NIFTY 50

Candidate: Dhruv | Role: Quant Engineer Intern | Evaluation: AlgoChowk Quantitative Challenge | Period: 2007–2026 (4,664 Trading Days)

1. Problem Formulation & Core Research Question

We empirically evaluate the hypothesis: *'After a significant one-day fall in NIFTY, the market tends to recover over the next few trading days.'* A common pitfall in market anomaly research is treating post-shock positive returns as an actionable alpha signal while ignoring the unconditional upward drift of equity indices. Over 19 years (September 2007 to September 2026, encompassing 4,664 audited daily bars), NIFTY 50 compounding produced a positive forward drift across every arbitrary rolling holding window. Our objective is to test whether post-drop forward returns exhibit genuine **Abnormal Returns (AR)** exceeding the unconditional distribution, evaluate execution feasibility across the intraday/overnight boundary, and establish empirical recovery hazard rates.

2. Experimental Design, Invariant Verification & Execution Reality

Data Sourcing & Invariants: Continuous daily OHLCV for NIFTY 50 (^NSEI) was extracted and audited through an invariant validation engine. Zero duplicate calendar dates, negative prices, or volume anomalies were identified. All bars satisfy strict envelope constraints: $High \geq \max(Open, Close)$ and $Low \leq \min(Open, Close)$.

Event Threshold (θ): We define a left-tail shock as a single-day close-to-close drop $\leq -2.0\%$ (representing the 95th percentile left-tail dislocation, $N = 200$ occurrences across 4,664 days). Robustness is cross-checked across a grid: $\theta \in \{-1.0\%, -1.5\%, -2.0\%, -2.5\%, -3.0\%\}$.

Eliminating Look-Ahead Bias: Naive event studies assume trade execution at Day T Close using Day T Close signals—a microstructure impossibility unless executed via Market-On-Close (MOC) facilities. We test two discrete models: **Model A (Day T Close)** benchmarks theoretical MOC capture with 2 bps execution slippage. **Model B (Day T+1 Open)** enforces realistic 9:15 AM execution, explicitly exposing the strategy to overnight gap risks.

3. Empirical Results: Event Forward Returns vs Unconditional Baseline

To test whether post-drop returns are distinct from random market drift, we calculate the unconditional baseline return across all 4,659 rolling 5-day windows. We apply two-sample Welch's t-tests (allowing unequal variances) and a 10,000-sample Circular Block Bootstrap (preserving 5-day return autocorrelation).

Horizon	Baseline Mean	Model A Mean	Abnormal Mean	Welch t-stat	Welch p-val	Block Boot p	H0 Rejected? ($\alpha=0.05$)	Model B Mean
1-Day (T+1)	+0.04%	+0.11%	+0.06%	+0.35	0.7291	0.4804	No ($p > 0.05$)	-0.04%
2-Day (T+2)	+0.09%	+0.09%	+0.00%	+0.01	0.9946	0.9925	No ($p > 0.05$)	-0.06%
3-Day (T+3)	+0.13%	+0.28%	+0.15%	+0.47	0.6405	0.5426	No ($p > 0.05$)	+0.12%
5-Day (T+5)	+0.21%	+0.32%	+0.10%	+0.25	0.8031	0.7863	No ($p > 0.05$)	+0.16%
10-Day (T+10)	+0.42%	+0.60%	+0.18%	+0.34	0.7364	0.7546	No ($p > 0.05$)	+0.44%

4. Survival Analysis: Empirical Hazard Rate of Price Recovery

Static holding returns obscure whether the market ever rebounded during the holding window. We model recovery dynamically as an empirical hazard function: what is the probability that NIFTY reclaims 50% or 100% of the single-day drop within K trading days?

Elapsed Window (K)	Prob($\geq 50\%$ Rebound)	Prob(100% Full Rebound)	Residual Dislocation Rate	Median Recovery Horizon
Day 1 (T+1)	43.5%	14.5%	85.5%	Unresolved (>1d)
Day 2 (T+2)	59.0%	26.0%	74.0%	Unresolved (>2d)
Day 3 (T+3)	65.5%	39.0%	61.0%	Median 50% Target Reclaimed
Day 5 (T+5)	72.0%	49.0%	51.0%	Full Recovery Remains <50%
Day 10 (T+10)	81.5%	65.5%	34.5%	Median Full Recovery Reclaimed
Day 20 (T+20)	85.5%	74.5%	25.5%	Long-Term Structural Regime Drift

Empirical Insight: By Day 5, the probability of full recovery is only **49.0%**. More than half of all significant shock events remain unrecovered after an entire trading week, directly disproving immediate deterministic bounce theories.

5. Critical Falsification, Regime Decomposition & Multiplicity Penalties

To vigorously challenge our findings, we decompose the shock events across three quantitative falsification dimensions:

1. The 200 SMA Market Regime Trap: Over **73% of large one-day falls (146 out of 200)** occur when NIFTY is trading below its 200-day Simple Moving Average. In structural bear regimes, one-day falls trigger strong **downside momentum and volatility clustering** (GARCH leverage effects), collapsing 3-day recovery to +0.14% (52.7% win rate). Conversely, in bull regimes (Price > 200 SMA, N=54), liquidity shocks mean-revert aggressively, averaging +0.65% over 3 days (59.3% win rate). Unconditioned dip-buying fails because it is overwhelmingly dominated by bear market falling knives.

2. Overnight Gap Decay: The mean overnight gap from Day T Close to Day T+1 Open is +0.15% (58.0% positive gap rate). When entering realistically at next morning's open, the 5-day forward return collapses from +0.32% to +0.16% (Abnormal Return = -0.06%, Welch t = -0.15, p = 0.8836). The marginal edge is entirely locked in the overnight gap and inaccessible to intraday traders.

3. Multiplicity & Holm-Bonferroni Correction: Testing across thresholds ($\theta \in [-1.0\%, -3.0\%]$) yields raw p-values between 0.2396 and 0.8031. After Holm-Bonferroni step-down penalties, zero parameter configurations achieve significance ($\alpha=0.05$).

Regime / Conditioning Filter	Sample Size (N)	3-Day Mean Return	5-Day Mean Return	Win Rate (5d)	Economic Interpretation
All Events (Unconditioned)	200	+0.28%	+0.32%	53.5%	Baseline market drift; statistically insignificant
Bull Regime (Price > 200 SMA)	54	+0.65%	+0.59%	53.7%	True liquidity shock absorption in structural uptrend
Bear Regime (Price < 200 SMA)	146	+0.14%	+0.21%	53.4%	Downside momentum & volatility clustering (Falling Knife)
Volume Climax (>1.5x 20d Mean)	15	-0.42%	-1.34%	40.0%	Institutional panic distribution; continued selling
Pin-bar Absorption ((C-L)/(H-L)≥0.35)	25	+1.12%	+1.57%	60.0%	Intraday buyer support defending daily lows

6. Realistic Event-Driven Backtest with Indian Market Friction

We simulate a portfolio starting with 1,000,000 INR from 2007 to 2026. Realistic Indian statutory costs are deducted: Securities Transaction Tax (STT), NSE turnover charges, SEBI fees, GST, and 4 bps round-trip bid-ask slippage (**0.07% total friction per trade**). Overlapping trades are locked out to prevent cash double-counting (124 executed trades).

Strategy Model	Execution Timing	Total Net Return	CAGR	Max Drawdown	Win Rate	Profit Factor
Model A (Blind 5d Holding)	Day T Close (MOC)	-7.16%	-0.39%	-46.18%	50.8%	0.96
Model B (Blind 5d Holding)	Day T+1 Open (9:15 AM)	-16.71%	-0.96%	-45.75%	49.2%	0.90
Risk-Managed (1.5x ATR Stop + TP)	Day T Close (Dynamic)	+0.82%	+0.04%	-52.38%	54.4%	1.02
NIFTY 50 Buy & Hold (Benchmark)	Passive Index Compounding	+482.4%	+9.72%	-59.90%	56.0%	1.42

7. Final Quantitative Conclusion & Institutional Recommendation

Empirical Verdict: The data **firmly rejects the unconditioned mean-reversion hypothesis**. Post-drop forward returns are statistically indistinguishable from random market drift (Welch t = +0.25, p = 0.8031; Bootstrap p = 0.7863). The apparent rebound is an optical illusion resulting from structural market drift, is economically captured by overnight gaps, and is destroyed by bear market momentum and transaction frictions.

Quant Recommendation: To construct a tradable alpha model, quants must: (1) filter strictly for **bull regimes (Price > 200 SMA)** to avoid bear momentum, (2) require **lower shadow pin-bar absorption** ((Close-Low)/(High-Low) ≥ 0.35) confirming intraday buyer defense, and (3) enforce **dynamic 1.5x ATR trailing stops** rather than blind multi-day holding.

Report reproduced deterministically from nifty50_daily.csv via algochowk-quant-research test suite (100% pass rate).